

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
 in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9729/02

Paper 2 Structured Questions

27 August 2018

Candidates answer on the question paper.

2 hours

Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group on all the work you hand in.
 Write in dark blue or black pen.

You may use pencil for any diagrams, graphs or rough working.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.

You are advised to show all working in calculations.
 You are reminded of the need for good English and clear presentation in your answers.
 You are reminded of the need for good handwriting.
 Your final answers should be in 3 significant figures.

You may use a calculator.

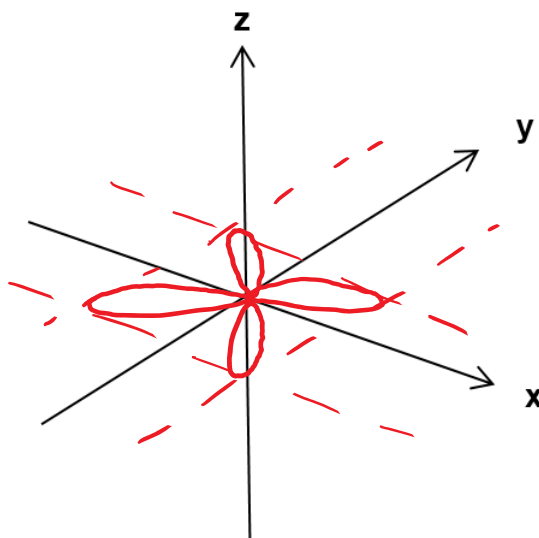
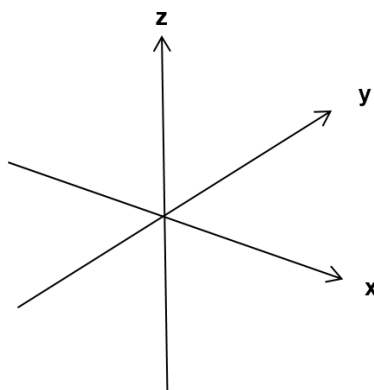
The number of marks is given in brackets []
 at the end of each question or part question.

At the end of the examination, fasten all your work
 securely together.

For Examiner's Use	
1	9
2	8
3	6
4	15
5	13
6	12
7	12
Significant Figures and Units	
Handwriting and Presentation	
Total	75

This document consists of **19** printed pages and **1** blank page.

1	(a)	The following diagram shows the second ionisation energy of some elements.																					
		<table><caption>Second Ionisation Energy Data (kJ mol⁻¹)</caption><thead><tr><th>Element</th><th>Second Ionisation Energy (kJ mol⁻¹)</th></tr></thead><tbody><tr><td>O</td><td>3400</td></tr><tr><td>S</td><td>2250</td></tr><tr><td>Cr</td><td>1600</td></tr><tr><td>Mn</td><td>1500</td></tr><tr><td>As</td><td>1800</td></tr><tr><td>Se</td><td>2050</td></tr><tr><td>Br</td><td>2000</td></tr><tr><td>Kr</td><td>2350</td></tr><tr><td>Te</td><td>1800</td></tr></tbody></table>		Element	Second Ionisation Energy (kJ mol⁻¹)	O	3400	S	2250	Cr	1600	Mn	1500	As	1800	Se	2050	Br	2000	Kr	2350	Te	1800
Element	Second Ionisation Energy (kJ mol⁻¹)																						
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Se	2050																						
Br	2000																						
Kr	2350																						
Te	1800																						
		(i)	Explain why the second ionisation energies of elements O, S, Se and Te show a decreasing trend. [1]																				
			The <u>number of principal quantum shells increases</u> / The valence electrons are <u>further away</u> [✓] <u>It is less strongly attracted to the nucleus</u> OR <u>Less energy is required to remove the valence electrons</u> [✓], hence second ionisation energy decreases.																				
		(ii)	Explain why the second ionisation energy of Br is lower than that of Se. [1]																				
			Se ⁺ : [Ar]4s ² 4p ³ Br ⁺ : [Ar]4s ² 4p ⁴ The <u>paired 4p/np electrons</u> [✓] / 2 electrons in the same 4p orbital in Br ⁺ experience <u>interelectronic repulsion</u> [✓], less energy is required to remove an electron from this orbital.																				
		(iii)	Write the full ground state electronic configuration for Cr. [1]																				
			1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁵ 4s ¹ [1]																				
		(iv)	Draw the d _{xy} orbital in the axes provided below. [1]																				



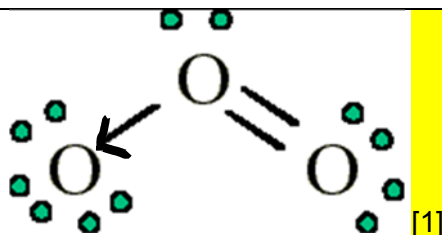
[1]

(b) Oxygen–oxygen bond lengths in some molecules are given below:

Molecule	Bond Length
Oxygen, O_2	0.121 nm
Hydrogen peroxide, H_2O_2	0.149 nm
Ozone, O_3	0.128 nm

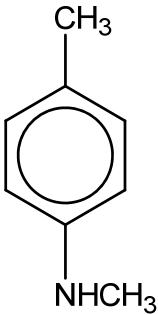
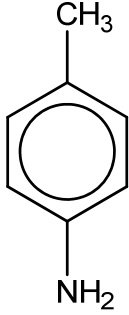
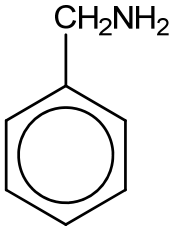
(i) Draw the structure of the molecule ozone, O_3 .

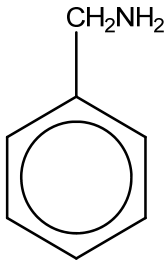
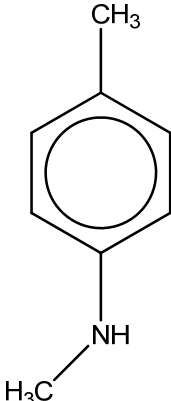
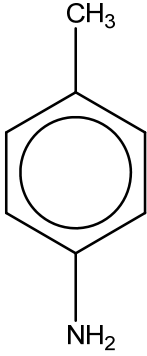
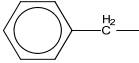
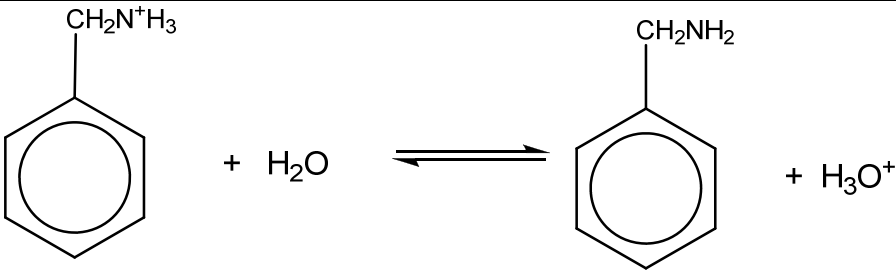
[1]



[1]

		(ii)	With reference to the data given above, comment and explain the oxygen–oxygen bond length in ozone as compared to oxygen and hydrogen peroxide. [2]
			Oxygen–oxygen bond length in ozone is longer than O=O in oxygen but shorter than O-O in hydrogen peroxide. [1] P orbital of oxygen overlap with π orbital of adjacent oxygen-oxygen double bond. OR Lone pair on oxygen atom can delocalised into the π electron cloud of adjacent oxygen- oxygen double bond [1]
		(iii)	Explain clearly why hydrogen peroxide is a liquid while oxygen is a gas at room conditions in terms of structure and bonding. [2]
			Both hydrogen peroxide and oxygen have simple molecular structure. [1] Oxygen have instantaneous dipole induced dipole forces of attraction between molecules [1] while hydrogen peroxide have hydrogen bonding between the molecules. [1] More energy is needed to overcome the stronger hydrogen bond [1], hence it has stronger intermolecular forces of attraction and exist as a liquid.
			[Total: 9]

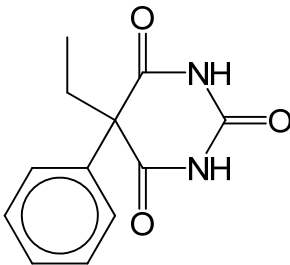
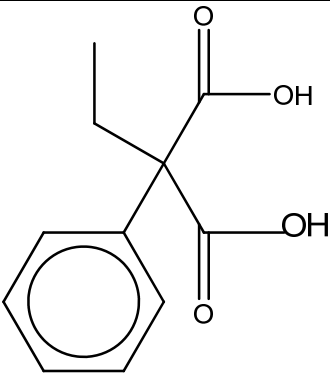
2	Compounds containing nitrogen are important to life and have applications in science and medicine. Three nitrogeneous bases have the following structures.  N,4-dimethylbenzenamine  4-methylphenylamine  benzylamine		
(a)	(i)	Arrange the three bases above in increasing order of pK_b. [1] Lowest pK_b $\longrightarrow$ Highest pK_b	

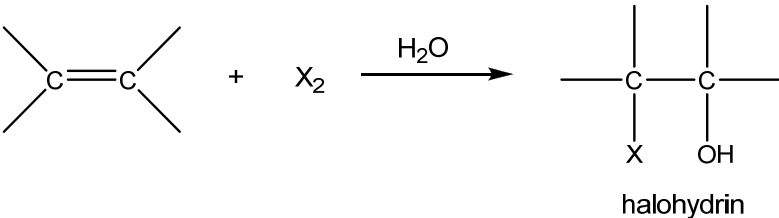
			 benzylamine  <i>N</i>,4-dimethyl benzenamine  4-methylphenylamine [1]
		(ii)	Explain in terms of their molecular structures why benzylamine and 4-methylphenylamine have different pK_b values. [2] In benzylamine, the presence of <u>electron donating</u>  <u>increases the electron density on N</u> ✓, making the <u>lone pair of electron on N more available</u> ✓ to accept a proton. Hence, benzylamine is the most basic. In 4-methylphenylamine, the <u>lone pair of electron on N delocalised into the π orbital of the benzene ring, decreasing the electron density on N</u> ✓, making the <u>lone pair of electron on N less available</u> ✓ to accept a proton. Hence, 4-methylphenylamine is the least basic.
		(iii)	Outline how <i>N</i>,4-dimethylbenzenamine may be produced from 4-methylphenylamine. [1] Heat 4-methylphenylamine with (ethanolic) CH_3Cl (in a sealed tube). [1]
	(b)		The pK_b of benzylamine is 4.66. 25.0 cm³ of 0.025 mol dm⁻³ benzylamine was completely neutralised by dilute hydrochloric acid of the same concentration. The salt formed reacts with water and the pH of the resultant solution is less than 7.
		(i)	Write the equation to show the reaction between the salt formed and water. [1] 

		(ii)	With reference to your equation in (b)(i), write an expression for the acid dissociation constant of the salt. [1]
			$K_a = \frac{\left[\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2 \right] \left[\text{H}_3\text{O}^+ \right]}{\left[\text{C}_6\text{H}_5\text{CH}_2\text{NH}_3^+ \right]}$ [1]
		(iii)	Hence, determine the pH of the resultant solution. [2]
			$[\text{salt}] = (6.25 \times 10^{-4}) \div (50/1000)$ $= 0.0125 \text{ mol dm}^{-3}$ ✓ $K_a = K_w / K_b$ $= \frac{10^{-14}}{10^{-4.66}}$ $= 4.571 \times 10^{-10}$ ✓ $[\text{H}^+] = \sqrt{K_a \times [\text{salt}]}$ $= \sqrt{4.571 \times 10^{-10} \times 0.0125}$ $= 2.390 \times 10^{-6}$ ✓ $\text{pH} = 5.62$ ✓
			[Total: 8]

3	Methanal is a colorless, strong-smelling gas used in making building materials and many household products. The Strecker synthesis is a route to preparing amino acids. Glycine, 2-aminoethanoic acid, can be prepared from methanal in this way. This is shown in the four steps reaction scheme below.
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reaction 1 reaction 2 reaction 3 reaction 4 2-aminoethanoic acid		
(a)	(i)	What is the role of ammonia, in the context of this synthesis? [1]
		Nucleophile [1] Allow Nucleophilic or Lewis base or Lone-pair donor
	(ii)	Compound E has the molecular formula CH_5NO . Suggest a structure for compound E . [1]
		 [1]
	(iii)	State the type of reaction for reaction 4. [1]
		Acidic hydrolysis [1]
(b)	The amino acid shown below is isoleucine, 2-amino-3-methylpentanoic acid. Molecule F can be used as the starting material to prepare this amino acid using a Strecker synthesis. Draw the skeletal structure of F. [1]	
		 [1]

	(c)	An amide bond is formed when two amino acids react together. Phenobarbital, which is a medication used to treat epilepsy, also has an amide bond in it.  phenobarbital Predict all the products formed when phenorbarbital undergoes acidic hydrolysis.	
		 [1] , CO₂ ☒, NH₄⁺ ☒	
			[2]
			[Total: 6]

4	(a)	Formation of 1,2-halo alcohols, also known as halohydrins, occurs via the addition reaction between an alkene and a halogen in the presence of water.  halohydrin where X = Cl or Br In a series of experiments, the reaction between propene and aqueous bromine was carried out with different concentrations of the two reagents, and the following relative initial rates were obtained.	
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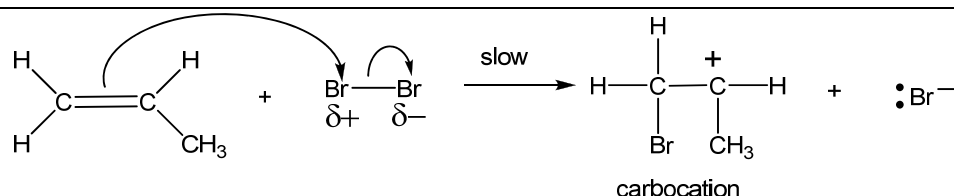
Experiment	[CH ₃ CH=CH ₂] / mol dm ⁻³	[Br ₂] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
1	0.020	0.020	1.00 x 10 ⁻³
2	0.030	0.020	1.50 x 10 ⁻³
3	0.040	0.030	3.00 x 10 ⁻³

(i)	Use these data to deduce the order of reaction with respect to each of the two reagents, showing how you arrive at your answers. [2]		
	Let the rate equation be: Rate = k[CH₃CH=CH₂]^x[Br₂]^y <table> <tr> <td> Comparing experiments 1 & 2, $\frac{\text{Rate}_1}{\text{Rate}_2} = \frac{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}$ $\frac{1.00 \times 10^{-3}}{1.50 \times 10^{-3}} = \frac{k [0.020]^x [0.020]^y}{k [0.030]^x [0.020]^y}$ x = 1 [1m] </td><td> Comparing experiments 1 & 3, $\frac{\text{Rate}_1}{\text{Rate}_3} = \frac{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}$ $\frac{1.00 \times 10^{-3}}{3.00 \times 10^{-3}} = \frac{k [0.020]^x [0.020]^y}{k [0.040]^x [0.030]^y}$ y = 1 [1m] </td></tr> </table>	Comparing experiments 1 & 2, $\frac{\text{Rate}_1}{\text{Rate}_2} = \frac{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}$ $\frac{1.00 \times 10^{-3}}{1.50 \times 10^{-3}} = \frac{k [0.020]^x [0.020]^y}{k [0.030]^x [0.020]^y}$ x = 1 [1m]	Comparing experiments 1 & 3, $\frac{\text{Rate}_1}{\text{Rate}_3} = \frac{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}{k [\text{CH}_3\text{CH}=\text{CH}_2]^x [\text{Br}_2]^y}$ $\frac{1.00 \times 10^{-3}}{3.00 \times 10^{-3}} = \frac{k [0.020]^x [0.020]^y}{k [0.040]^x [0.030]^y}$ y = 1 [1m]
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(ii)	Hence write a rate equation for the reaction. [1]		
	Rate = k[CH ₃ CH=CH ₂][Br ₂]		
(iii)	Calculate the rate constant for the reaction, giving its units. [2]		
	Using experiment 1, $k = \frac{\text{Rate}}{[\text{CH}_3\text{CH}=\text{CH}_2][\text{Br}_2]}$ $k = \frac{1.00 \times 10^{-3}}{[0.020][0.020]} = 2.50 [1m] \text{ mol}^{-1}\text{dm}^3\text{s}^{-1} [1m]$		
(iv)	State and explain how the rate of reaction may change if chlorine is used instead of bromine in the reaction with propene. [1]		
	Rate will decrease [✓] because Cl-Cl bond is stronger than Br-Br [✓].		
(b) (i)	The mechanism of the addition reaction between propene and aqueous Br₂ involves three steps. There is an initial attack by the π electron pair of the alkene on Br₂ to yield a carbocation intermediate.		

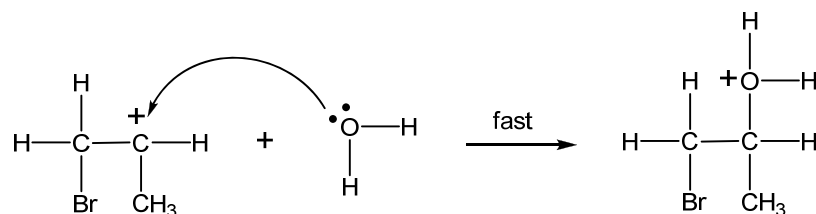
- This is followed by the nucleophilic attack of the lone pair of electrons on oxygen in water on the carbocation intermediate.
- The third step involves the loss of H^+ ion which then yields the neutral bromohydrin.

Using the information given above, describe a mechanism for this reaction.

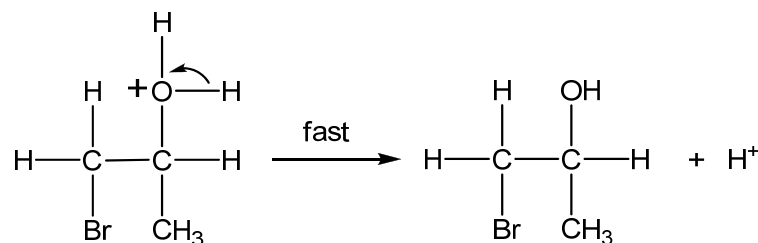
[3]



Dipoles [✓], two curly arrows [✓], correct carbocation [✓], label slow step [✓], Br^- [✓]



Lone pair + curly arrow [✓], intermediate [✓], label fast step [✓]



Curly arrow [✓], final organic product [✓], H^+ [✓]

11 [✓] – 3m

6 to 10 [✓] – 2m

3 to 5 [✓] – 1m

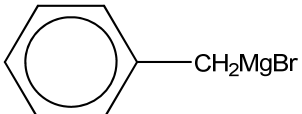
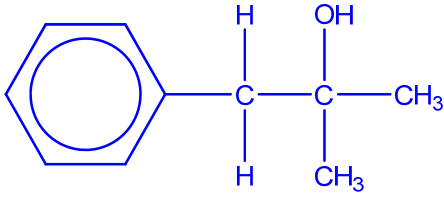
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(ii)

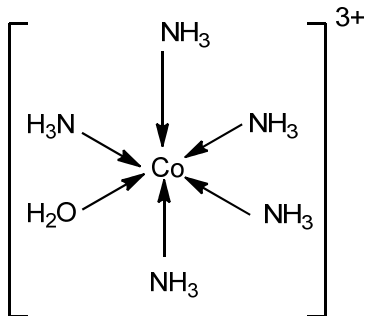
Based on your mechanism drawn, explain whether it is consistent with the rate equation proposed. [1]

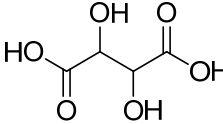
It is **consistent** as there is **1 propene molecule and 1 Br_2 molecule in the rate determining step**, which is consistent with the rate equation.

Or

			To allow ecf if rate equation is wrong
	(c)	Grignard reagents are organo-magnesium halides, commonly used in synthesis to prepare a variety of organic compounds. The carbon-magnesium bonds in Grignard reagents are highly polar and this makes it extremely useful in organic synthesis as it is able to react with other polar organic molecules to form carbon-carbon bonds. An example of the use of a Grignard reagent is the two-step reaction of $\text{CH}_3\text{CH}_2\text{MgBr}$ with butanone, $\text{CH}_3\text{CH}_2\text{COCH}_3$, to form 3-methylpentan-3-ol. $\begin{array}{c} \text{CH}_3\text{CH}_2\text{MgBr} + \text{CH}_3\text{CH}_2\text{COCH}_3 \xrightarrow{\text{step 1}} \begin{array}{c} \text{OMgBr} \\ \\ \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{CH}_2\text{CH}_3 \end{array} \\ \text{Grignard reagent} \end{array}$ $\begin{array}{c} \downarrow \text{step 2} \quad \text{water} \\ \begin{array}{c} \text{OH} \\ \\ \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{CH}_2\text{CH}_3 \end{array} + \text{Mg(OH)Br} \\ \text{3-methylpentan-3-ol} \end{array}$	
	(i)	Suggest the type of reaction that has taken place in step 1.	[1]
		Nucleophilic addition [1]	
	(ii)	Suggest the structural formula of the final organic product formed when  is reacted with propanone, CH_3COCH_3, in a similar two-step process.	
			[1]

		(iii)	Suggest a suitable Grignard reagent and another organic compound to be used if propan-2-ol is to be prepared using a similar two-step process. [2]
			CH_3MgBr [1] CH_3CHO [1]
		(iv)	The Grignard reagent $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ can be readily converted into a carboxylic acid by using carbon dioxide. Suggest the structural formula for the organic product formed. [1]
			$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
			[Total: 15]

5	(a)	Cobalt forms many coloured complexes. Cobalt(III) chloride combines with ammonia to form a pink coloured compound A , $\text{CoCl}_3 \cdot \text{H}_2\text{O} \cdot 5\text{NH}_3$ ($M_r = 268.4$) in which the coordination number of cobalt is 6. 1.00 g of A is dissolved in 25 cm^3 of water and the solution is titrated with $0.500 \text{ mol dm}^{-3}$ silver nitrate solution. It is found that 22.40 cm^3 of silver nitrate is required for complete reaction.	
		(i)	Calculate the number of moles of free chloride ions per mole of A .
			no. of moles of $\text{AgNO}_3 = 0.500 \times (22.40/1000) = 1.12 \times 10^{-2} \text{ mol}$ [1] no. of moles of $\text{Cl}^- = 1.12 \times 10^{-2} \text{ mol}$ no. of moles of A = $1.00/268.4 = 3.73 \times 10^{-3} \text{ mol}$ no. of moles of free Cl^- ions per mole of A = $(1.12 \times 10^{-2}) / (3.73 \times 10^{-3}) = 3$ [1] [2]
		(ii)	Draw the structure of the complex ion in A .
			 [1]
		(iii)	When the pink compound A is heated, water vapour and ammonia were evolved to give a purple solid B . A and B have the same coordination number. Suggest the formula of the complex in the purple solid B .
			$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ or $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ [1]

			[1]
		(iv)	Account for the difference in the colour of A and B .
			The different ligands in A and B cause the energy gap between the split d-orbitals to be different . [1] The wavelength of light absorbed is different [1] for A and B and thus the colour observed are different. [2]
		(b)	Aqueous hydrogen peroxide is fairly stable, but when a mixture of a cobalt(II) salt and tartaric acid is added to aqueous hydrogen peroxide, the initially pink solution slowly turns into a green cobalt(III) species.  tartaric acid After a while, oxygen gas is vigorously evolved and the solution turns back to pink again.
		(i)	State the role of the cobalt(II) salt in this reaction and support your answer by referring to the observations.
			Role of the cobalt(II) salt: catalyst [1] Observation with explanation: - The oxygen gas is vigorously evolved showing that the reaction speeds up [1] OR - The solution turns from pink to green to pink (WTTE) showing that Co²⁺ is regenerated [1] [2]
		(ii)	Tartaric acid acts as a complexing agent in this reaction to stabilise the Co³⁺ cation. With the aid of relevant data from the <i>Data Booklet</i>, show that Co³⁺ is not stable in aqueous solution. $\text{Co}^{3+} + \text{e} \rightleftharpoons \text{Co}^{2+} \quad E_{\text{red}} = +1.89\text{V}$ $\text{O}_2 + 4\text{H}^+ + 4\text{e} \rightleftharpoons 2\text{H}_2\text{O} \quad E_{\text{red}} = +1.23\text{V}$ $E^\circ_{\text{cell}} = +1.89 - (+1.23) = +0.66\text{ V}$ [1] Co³⁺ oxidises water readily to form Co²⁺ / Co³⁺ is readily reduced by water to form Co²⁺. [1] [2]
		(c)	A student wanted to measure the standard cell potential, E°_{cell} , between the Co ²⁺ /Co half-cell and the Fe ³⁺ /Fe ²⁺ half-cell. She set up and connected the two half cells and obtained a reading.
		(i)	Calculate the value E°_{cell} that will be obtained by the student.
		[R] [O]	$\text{Fe}^{3+} + \text{e} \rightleftharpoons \text{Fe}^{2+} \quad E_{\text{red}} = +0.77\text{V}$ $\text{Co}^{2+} + 2\text{e} \rightleftharpoons \text{Co} \quad E_{\text{red}} = -0.28\text{V}$

			$E^\ominus_{\text{cell}} = +0.77 - (-0.28) = +1.05 \text{ V}$ [1]
		(ii)	State and explain what happens to the standard cell potential, $E^\ominus_{\text{cell}}$ when ammonia is added to the Co^{2+}/Co half-cell.
			When NH_3 is added, the following equilibrium is set up in the Co^{2+}/Co half-cell: Either: $[\text{Co}(\text{NH}_3)_6]^{2+} + 2\text{e}^- \rightleftharpoons \text{Co} + 6\text{NH}_3$ $E_{\text{red}} = -0.43\text{V}$ [1] (Give B.O.D. if the value -0.43V is not quoted in the answer) OR $[\text{Co}^{2+}]$ decreases as $\text{Co}(\text{OH})_2(\text{s})$ is formed. [1] the E_{ox} of Co^{2+}/Co will become more negative/ position of the equilibrium $\text{Co}^{2+} + 2\text{e}^- \rightleftharpoons \text{Co}$ will shift left Hence, the $E^\ominus_{\text{cell}}$ becomes more positive. [1]
			[2]
			[Total: 13]

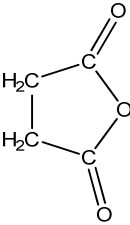
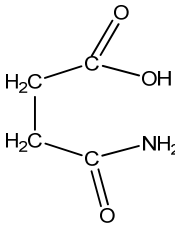
6	A student plotted the sketch graph below to show how the entropy of a sample of H_2O varies with temperature.		
	(a)	Identify the state of H_2O in Phase 2 and 3 respectively. Suggest a value of T_2 .	
		In phase 2: In phase 3: value of T_2 : [1]	
		In phase 2: liquid In phase 3: gas T_2 : 373K	
	(b)	Suggest why entropy of H_2O is zero at 0 K.	
		 [1]	
		It is at maximum state of order or no disorder or perfectly ordered or pure crystalline structure.	

	(c)	Explain why the entropy change, ΔS , at temperature T_2 is much larger than that at temperature T_1 .								
		[2]								
		At T_2, the increase from phase 2 to 3 is much larger as the <u>gas molecules</u> in phase 3 have <u>more ways in which the particles can be arranged</u> than particles in liquid. [1] At T_1, the particles in phase 2 (liquid) are still orderly arranged compared to the ordered arrangement of particles in phase 1. [1] Thus, the increase in disorder is smaller from phase 1 to 2. (2nd marking point: mention of smaller increase from phase 1 to 2 as it is from an ordered arrangement to slightly disordered arrangement in liq.)								
	(d)	It requires 3.49 kJ of heat energy to convert 1.53 g of H_2O from the state in phase 2 to phase 3 at temperature T_2 and 100kPa. Use these data and your value of T_2 in part (a) to calculate the value of ΔS, including units, for the conversion of one mole of H_2O from the state in phase 2 to phase 3 at temperature T_2.								
		[3]								
		Amt of water = $\frac{1.53}{18} = 0.0850$ enthalpy change reaction per mole of water = $\frac{3.49}{0.0850} = +41.1 \text{ kJ mol}^{-1}$ [1] award if + sign is omitted, but penalise negative sign is included $\Delta G = \Delta H - T\Delta S = 0$ $+41.1 - (373)\Delta S = 0$ equate $\Delta G = 0$, substitute ΔH, T_2 into equation [1] ecf $\Delta S = 0.110 \text{ kJ K}^{-1} \text{ mol}^{-1}$ correct computation (allow ecf) and units [1] ecf								
	(e)	The student wants to find out if dissolving a salt, silver fluoride in water is always a spontaneous process. He must first find the enthalpy change of solution of silver fluoride in water. Some enthalpy changes for silver fluoride are shown in the table. <table><tr><td></td><td>$\Delta H / \text{kJ mol}^{-1}$</td></tr><tr><td>Lattice energy of silver fluoride</td><td>-950</td></tr><tr><td>enthalpy change of hydration for silver ions</td><td>-464</td></tr><tr><td>enthalpy change of hydration for fluoride ions</td><td>-506</td></tr></table>		$\Delta H / \text{kJ mol}^{-1}$	Lattice energy of silver fluoride	-950	enthalpy change of hydration for silver ions	-464	enthalpy change of hydration for fluoride ions	-506
	$\Delta H / \text{kJ mol}^{-1}$									
Lattice energy of silver fluoride	-950									
enthalpy change of hydration for silver ions	-464									
enthalpy change of hydration for fluoride ions	-506									
	(i)	Use the data provided to calculate a value for the enthalpy change of solution of silver fluoride in water.								

			[2]
		$\Delta H_{\text{solution}} = \Delta H_{\text{hyd}} \text{Ag}^+ + \Delta H_{\text{hyd}} \text{F}^- - \text{LE}$ $= (-464) + (-506) - (-950) \quad [1] \text{ working}$ $= -20 \text{ kJ mol}^{-1} \quad [1] \text{ answer}$	
		(ii) If entropy change for dissolving silver fluoride in water has a positive value, explain why dissolving of silver fluoride in water is always a spontaneous process.	
			[2]
		$\Delta G = \Delta H - T\Delta S$ Relate to signs of ΔS and ΔH [1]: Since ΔS is positive, the term $(-T\Delta S)$ is always negative. Thus, since ΔH is negative, <u>ΔG is always negative at all temperature.</u> [1]	
		(iii) Explain why the enthalpy change of hydration of the fluoride ions is more negative than the enthalpy change of hydration of the chloride ions.	
			[1]
		$\Delta H_{\text{hydration}} \propto \text{charge density} [\sqrt{v}]$ Fluoride (ions) are smaller (than chloride), with <u>higher charge density</u> $[\sqrt{v}]$	
			[Total: 12]

7	The uses of carboxylic acids are so extensive that they can be divided into several industries, such as pharmaceuticals or food among others.		
	(a)	Ethanol and ethanoic acid react reversibly to form ethyl ethanoate and water according to the equation: $\text{CH}_3\text{COOH(l)} + \text{CH}_3\text{CH}_2\text{OH(l)} \rightleftharpoons \text{CH}_3\text{COOCH}_2\text{CH}_3\text{(l)} + \text{H}_2\text{O(l)}$ A student mixed 8.00×10^{-2} mol of ethanoic acid and 1.20×10^{-1} mol of ethanol in a conical flask and the flask was sealed with a bung and allowed to reach equilibrium at 20°C. The equilibrium mixture is placed in a graduated flask and the volume made up to 250 cm^3 with distilled water. A 10.0 cm^3 sample of this equilibrium mixture was placed in a conical flask with a few drops of phenolphthalein and titrated with $0.100 \text{ mol dm}^{-3}$ of sodium hydroxide from a burette. The indicator turned pink when 6.40 cm^3 of NaOH had been added.	
	(i)	Calculate the amount of CH_3COOH in the equilibrium mixture. Amount of NaOH = $0.100 \times \frac{6.40}{1000} = 0.00064 \text{ mol}$ Amount of CH_3COOH in $10.0 \text{ cm}^3 = 0.00064 \text{ mol}$ Amount of CH_3COOH in $250 \text{ cm}^3 = \frac{250}{10} \times 0.00064 = 0.016 \text{ mol}$ [1]	

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	(c)	Malonic acid can undergo dehydration with P_4O_{10} to give a foul-smelling gas, A . At 30.5 kPa, 0.1057 g of A occupies 200 cm ³ at a temperature of 200°C. Determine the relative molecular mass of A .
		$pV = nRT$ $30.5 \times 10^3 \times \frac{200}{10^6} = \frac{0.1057}{M_r} \times 8.31 \times (200 + 273)$ [1] $M_r = 68.1$ (1 d.p) [1]
		[2]
	(d)	At high temperature, succinic acid can also undergo dehydration to produce a neutral compound B, $C_4H_4O_3$ which does not react with sodium metal or Brady's reagent. Compound B reacts with ammonia to give a compound C, $C_4H_7NO_3$, which reacts with cold NaOH(aq), but not with cold HCl(aq). Suggest structures for B and C and explain the observations.
		B does not react with Na; absence of –OH group. [✓] B does not react with Brady's reagent; absence of carbonyl group. [✓] C does not react with cold HCl(aq); absence of basic group [✓], amide likely to be present [✓]. C undergoes acid base with cold NaOH(aq) [✓]; it contains an acidic group (carboxylic acid). [✓] Max [2] for explanations  compound B  compound C [1] each
		[4]
		[Total: 12]